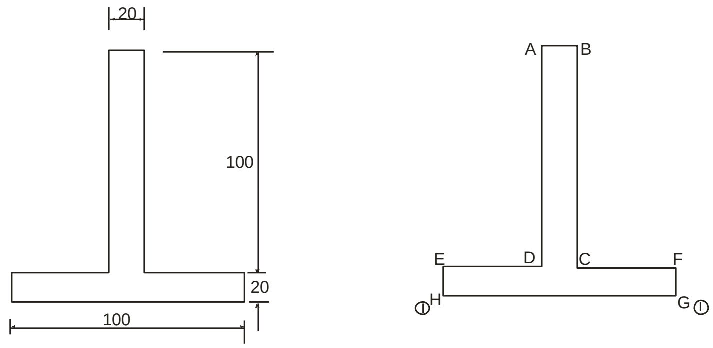
Problem 4.1 Find the centroid of the section shown in Fig. 4.10(a).

Solution: The section will be split up into two rectangular areas ABCD and EFGH as shown in Fig. 4.10(b) of areas,

$20 \times 100 = 2000$ mm^2

and $100 \times 20 = 2000$ mm^2. respectively



(a) (b)

Fig. 4.10

Centroidal distance of ABCD from the axis 1 – 1 = 70 mm

Centroidal distance of EFGH from the axis 1 – 1 = 10 mm

Let $\bar{y}$ be the height of the centroid of the section from the axis 1 – 1

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{2000 \times 70 + 2000 \times 10}{2000 + 2000} \text{ mm}$$

= 40 mm above the axis 1 – 1

The above computation may be conveniently worked out in a tabular form as shown below:

Component	Area A mm^2	Centroidal Distance From 1 – 1 y mm	Ay mm^2
ABCD EFGH	2000 2000	70 10	140000 20000
Total	4000		60000

$$\therefore \quad \bar{y} = \frac{\sum Ay}{\sum A} = \frac{160000}{4000} = 40 \text{ mm}$$

Pronlem 4.2 Find the centroid of the section in Fig. 4.11.

Solution: The given section will be split up into a number of components. The area of the various components and their centroidal distance from axis 1 – 1 and the moments of the individual components about the axis 1- 1 are shown in the following table:

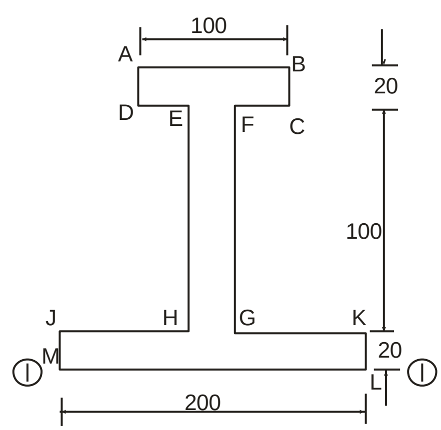


Fig. 4.11

Component	Area A mm^2	Centroidal Distance From y 1 – 1 mm	Ay mm^2
ABCD 100 × 20 EFGH 10 × 100 JKLM 200 × 20	2000 1000 4000	130 70 10	260000 70000 40000
Total	7000		370000

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{370000}{7000} \text{ mm} = 52.86 \text{ mm above the axis } 1 - 1.$$

Problem 4.3 Find the centroid of the section shown in Fig. 4.12.

Solution: The given section may be split up into two rectangles ABCD and EFGH as shown in

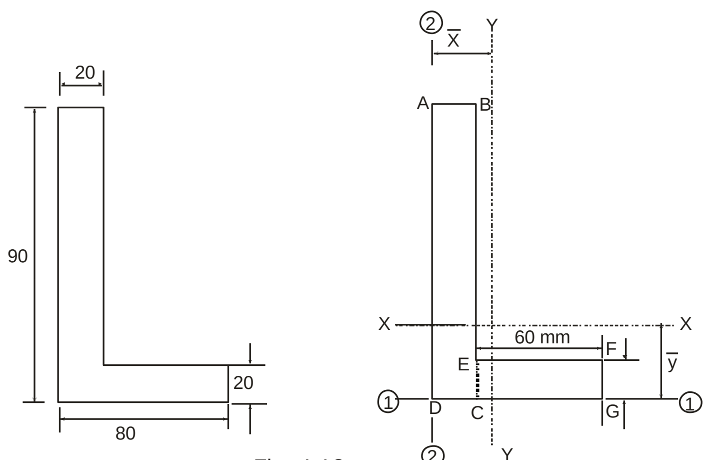


Fig. 4.12

Fig. 4.12. The position of the centroid of the section with respect to the axis 1 – 1 and 2 – 2 will now be worked out. The relevant computations are shown in the following table:

Component	Area A mm^2	Centroidal Distance y From 1 – 1 mm	Centroidal Distance x From 2 – 2 mm	Ay mm^3	Ax mm^3
ABCD 20 × 90 EFGC 60 × 20	1800 1200	45 10	10 50	81000 12000	18000 60000
Total	3000			93000	78000

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{93000}{3000} = 31 \text{ mm}$$

$$\bar{x} = \frac{\sum Ax}{\sum A} = \frac{78000}{3000} = 26 \text{ mm}$$

Problem 4.4 Find the moment of inertia of the section shown in Fig. 4.13 about the centroidal axis perpendicular to the web.

Solution: The given section will be split up into components ABCD, EFGH and JKLM. The areas of the individual components, their centroidal distances from the axis 1 – 1 through the bottom edge and the moments of inertia of the individual components about their respective centroidal axes parallel to the axis 1 – 1 are given in the following table:

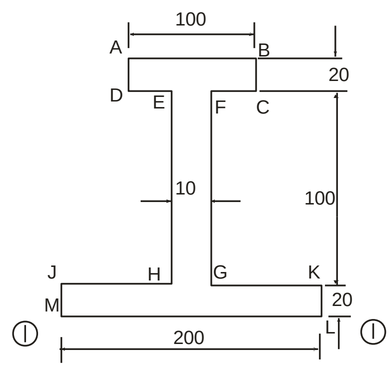


Fig. 4.13

Component	Area A mm^2	Centroidal distance y from 1 – 1 mm	Ay mm^3	Ay^2 mm^4	I_{self} mm^4
ABCD	2000	130	260000	33800000	$\frac{100\times 20^3}{12} = 66667$
EFGH	1000	70	70000	4900000	$\frac{10\times 100^3}{12} = 833333.33$
JKLM	4000	10	40000	400000	$\frac{200\times 20^3}{12} = 133333$
TOTAL	7000		370000	39100000	1033333.33

Distance of the centroidal axis XX from the axis 1- 1

$$= \bar{y} = \frac{\sum Ay}{\sum A} = \frac{370000}{7000} \text{ mm} = 52.86 \text{ mm}$$

Moment of inertia about the axis 1 – 1

$$= I_{1-1} = \sum I_{self} + \sum Ay^2$$

$$= 1033333.33 + 3910.0000 = 40133333.33 \text{ mm}^4$$

But $I_{1-1} = I_{xx} + \sum A\bar{y}^2$

$\therefore$ $40133333.33 = I_{xx} + 7000 \times 52.86^2$

$\therefore$ $I_{xx} = 20574076.13 \text{ mm}^4$

Problem 4.5 Find the moment of inertia about the centroidal axes XX and YY of the section shown in Fig. 4.14.

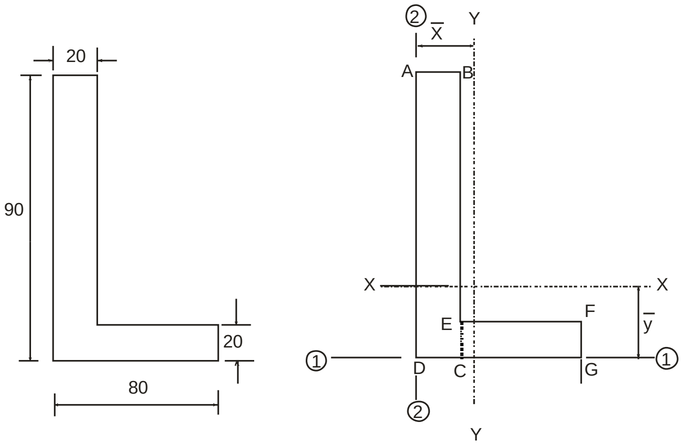


Fig. 4.14

Solution: To find the moment of inertia about the axis XX:

The following table shows the relevant computations:

Component	Area A mm^2	Centroidal distance y from 1 – 1 mm	Ay mm^3	Ay^2 mm^4	I_{self} mm^4
ABCD	1800	45	81000	3645000	$\frac{20\times90^3}{12}$ = 1215000
EFGH	1200	10	12000	120000	$\frac{60\times20^3}{12}$ = 40000
TOTAL	3000		93000	3765000	1255000

$\therefore \qquad \bar{y} = \frac{\sum Ay}{\sum A} = \frac{93000}{3000} = 31$

Moment of inertia about the axis 1 – 1 = $I_{1-1} = \sum I_{self} + \sum Ay^2$

But $\qquad I_{1-1}$ = 1255000+3765000 = 5020000 mm^4

5020000 = I_{xx} + 3000 × 312

$\therefore \qquad I_{xx}$ = 2137000 mm^4

To find the moment of inertia about the axis YY:

The following table shows the relevant computations:

Component	Area A mm^2	Centroidal distance x from 2 – 2	Ax mm^3	Ax^2 mm^4	I_{self} mm^4

		mm			
ABCD	1800	10	18000	180000	$\frac{90\times20^3}{12}$ = 60000
EFGH	1200	50	60000	3000000	$\frac{20\times60^3}{12}$ = 360000
TOTAL	3000		78000	3180000	420000

$\therefore \qquad \bar{x} = \frac{\sum Ax}{\sum A} = \frac{78000}{3000} = 26$ mm

Moment of inertia about the axis 2 – 2

$= I_{2-2} = \sum I_{self} + \sum Ax^2$

$\therefore \qquad I_{2-2} = 420000 + 3180000 = 3600000$

But $\qquad I_{2-2} = I_{yy} + (\sum A)(\bar{x}^2)$

$3600000 = I_{yy} + 3000 \times 26^{2\,-}$

$I_{yy} = 1572000$ mm^4

Problem 4.6 Calculate the moment of inertia about the horizontal and vertical gravity axes (I_{xx} and I_{yy}) of the section shown in Fig. 4.15.

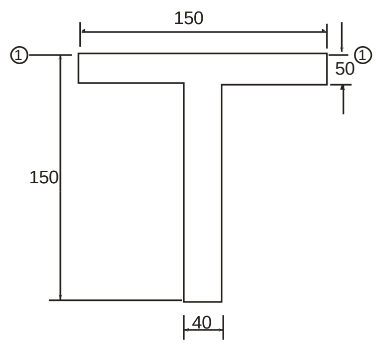


Fig. 4.15

Solution: The given section is split up into two components. The areas of the individual components, their centroidal distances from the axis 1 – 1 through the top edge and the Moments of Inertia of the individual components about their centroidal axes parallel to the axis 1 – 1 are tabulated below:

Component	Area A mm^2	Centroidal distance y from 1 – 1 mm	Ay mm^3	Ay^2 mm^4	I_{self} mm^4
Top Flange	7500	25	187500	4687500	$\frac{150\times50^3}{12}$ = 1562500
Web	4000	100	400000	40000000	$\frac{40\times100^3}{12}$ = 3333333.3
TOTAL	11500		587500	44687500	4895833.3

Distance of the centroidal axis XX from 1 – 1

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{587500}{11500} = 51.09 \text{ mm}$$

$$I_{1-1} = \sum I_{self} + \sum Ay^2 = 4895833.3 + 44687500 = 49583333.3$$

But $\quad I_{1-1} = I_{xx} + (\sum A)\bar{y}^2$

$$49583333.3 = I_{xx} + 11500 \times 51.09^2$$

$\therefore \quad I_{xx} = 19566170.15 \text{ mm}^4$

$$I_{yy} = \frac{50 \times 150^3}{12} + \frac{100 \times 40^3}{12} = 14595833.33 \text{ mm}^4$$

Problem 4.7 Find the moment of inertia about the centroidal axis XX for the section shown in Fig. 4.16. The dimensions shown in the Fig. 4.16 are in cm.

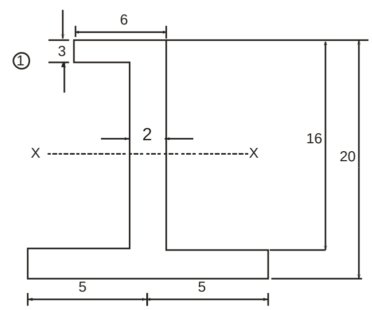


Fig. 4.16

Solution: The given section will be split up into three rectangular components, i.e., the top flange, the web and the bottom flange.

The properties of the components are tabulated below:

Component	Area of (cm^2)	Centroidal distance y from 1 – 1 (cm) bottom edge	Ax (cm^3)	Ax^2 (cm^4)	I_{self} (cm^4)
Top Flange	18	18.5	333	6160.50	$\frac{6\times 3^3}{12} = 13.5$
Web	26	10.5	273	2866.50	$\frac{2\times 13^3}{12} = 366.167$
Bottom flange	40	2	80	160	$\frac{10\times 4^3}{12} = 53.333$
TOTAL	84		686	9187	433

Distance of centroidal axis XX from the bottom edge 1 – 1

$$\bar{y} = \frac{\sum Ay}{\sum A} = \frac{686}{84} = 8.17 \text{ cm}$$

Moment of inertia about the axis 1 – 1

$= I_{1-1} = = \sum I_{self} + \sum Ay^2 = 433+9187 = 9620\ cm^4$

But $\quad I_{1-1} = I_{xx} + A\bar{y}^2$

$\therefore \quad 9620 = I_{xx} + 84 \times 8.172$

$\therefore \quad I_{xx} = 4015.38\ cm^4$

Exercise

4.1 Define centre of gravity and centroid.

4.2 Define radius of gyration and moment of inertia.

4.3 Find the expression for moment of inertia of a rectangular section about the axis passing through its base.

4.4 State the parallel axis theorem and perpendicular axis theorem.

4.5 Find the expression for moment of inertia of a hollow rectangular section about the axis passing through its base.

4.6 Find the centre of gravity and moment of inertia of the following sections about their centroidal axis: